

Joseph Kayijuka

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EDUCATION

University of Zielona Gora, Zielona Gora, Poland

Graduated: 15 July 2026

Bachelor of Science in Computer Science and Econometrics — Final GPA: 4.0/5

Relevant Coursework: Calculus, Linear Algebra, Probability, Statistics, Optimization, Mathematical Finance, Econometrics, Algorithms and Data Structures, Microeconomics, Macroeconomics, Accounting, Public Finance, Databases, Object-Oriented Programming

TU Chemnitz, Chemnitz, Germany

Erasmus Exchange

Wirtschaftsinformatik (Business Information Systems)

PROJECTS & EXPERIENCE

NLP-Powered Transaction Automation for a Fintech App

October 2025 – Present

Final Year Group Project

- Developing an Intent Management System (IMS) to parse natural language commands and automate complex savings transactions.
- Building a custom Named Entity Recognition (NER) model in Python to extract and structure financial entities (amount, frequency, goal) into a JSON output for backend systems.
- Aiming to significantly improve user experience by enabling hands-free, voice-or-text-driven automation of savings plans, directly impacting the core functionality of the application.

Python Development Intern

August 2025

Coherent Solutions

- Optimized legacy Python scripts for processing Finite Element Method (FEM) simulation data, achieving a significant performance increase by replacing loops with vectorized NumPy operations.
- Automated data-processing workflow to format and combine multi-axis simulation results using linear algebra, preparing the data for visualization and further analysis.

Simulation Model for Detecting Household Misreporting (Modeling Week 2025)

May 2025

University of Zielona Gora, Applied Math Society

- Engineered a Monte Carlo simulation in Python (NumPy, SciPy) to identify optimal water consumption thresholds for detecting household occupancy misreporting.
- Modeled the accuracy vs. false-positive trade-off to find an optimal threshold, achieving a simulated 84% detection rate with a 15% false-positive rate.
- Proposed a practical framework to apply the model to real-world datasets, designed to improve validation efficiency and potentially boost municipal revenue.

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, C++, Java

Quantitative & Data Analysis: Regression Modeling, Machine Learning, Forecasting, Statistical Analysis, Time Series, Optimization

Libraries & Tools: NumPy, Pandas, Scikit-learn, Statsmodels, PyTorch, FastAPI, Git, Docker, Tableau

Databases: MySQL, PostgreSQL

LANGUAGES

English (Fluent), French (Fluent), German (Intermediate)